

ANDRÉS DUARTE ORNELAS

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EDUCATION

University of Texas at San Antonio (UTSA)

San Antonio, TX

Master of Science in Information Technology, Cybersecurity Concentration

May 2026

- U.S. residency year (2025–2026) of the dual UTSA–Tecnológico de Monterrey Master's in Cybersecurity
- Coursework: Security Risk Analysis, Strategic Management of IT, Data Mining, Cyber Warfare, plus jointly-advised Independent Study research paper

Monterrey Institute of Technology (ITESM)

Monterrey, MX

Master's in Cybersecurity

Expected: December 2026

Bachelor of Science in Electronics Engineering, Specialization in Cybersecurity

June 2024

CERTIFICATIONS

Cisco CyberOps Associate Certification

November 2023

Cisco Networking Academy

- Demonstrated comprehensive understanding of Security Operations including intrusion detection, analysis, and network monitoring
- Completed 52 hands-on labs (30+ hours) using Security Onion, ELK Stack, and Cisco Packet Tracer

AWS Certified AI Practitioner

July 2026

Amazon Web Services (AWS)

- Validated foundational knowledge of AI, ML, and generative AI concepts, including model types, use case selection, and responsible AI practices
- Demonstrated proficiency with AWS AI/ML services (Amazon Bedrock, SageMaker, Amazon Q) and prompt engineering techniques for foundation models

PROFESSIONAL EXPERIENCE

Cybersecurity Trainee

August 2024 – December 2024

Dipole Digital, Monterrey, MX & Hermosillo, MX (Hybrid)

- Evaluated risks in production environments and developed tailored mitigation solutions
- Designed and implemented segmented office network using VLANs with DMZ for internet-facing services
- Configured traffic rules to control inter-segment communication and enhance network security
- Performed web vulnerability scanning using Zed Attack Proxy (ZAP) to identify and remediate threats
- Assisted in creating and implementing company-wide cybersecurity policy framework
- Gained hands-on experience in security incident management and response

Cybersecurity Engineering Intern

March 2024 – June 2024

Dipole Digital, Monterrey, MX (Remote)

- Conducted comprehensive vulnerability assessments and implemented system hardening strategies
- Documented security findings and provided actionable recommendations to enhance cybersecurity posture

RELEVANT ACADEMIC PROJECTS

Empirical Characterization of a Commercial Coding Agent for CLI-Based OSPF Fault Diagnosis

January 2026 – May 2026

- Conducted the first empirical evaluation of a general-purpose coding agent (Claude Code) performing autonomous OSPF fault diagnosis in a traditional, non-SDN CLI environment, addressing a gap in existing LLM network operations research
- Designed a 50-session quantitative experiment using a three-router GNS3/VyOS topology modeled after CCNA 11.6.3, with six simultaneously active faults per session and automated pre/post verification via Netmiko snapshots
- Achieved an effective fault resolution rate of ~97% and a perfect-run rate of ~78% at a mean cost of \$0.81/session; zero sessions introduced a configuration regression

- Publicly released experimental materials, router configurations, and analysis scripts at github.com/Doarte45/LLMnetopsLab

Cloud as a Service Commercial and Operative Design

August 2023 – October 2023

- Designed multi-cloud service architecture using AWS, Azure, and Google Cloud for company Quanti
- Developed commercial strategy for service delivery and operational framework

Digital Forensic Investigation: “Posthumous Digital Note” Case Analysis

August 2025

- Collaborated on comprehensive digital forensic analysis of suspicious death case using industry-standard tools (Autopsy/Sleuth Kit, Volatility, Registry Explorer)
- Analyzed disk images, RAM dumps, and Windows event logs to reconstruct timeline and establish evidence of third-party involvement
- Documented complete chain of custody following ISO/IEC 27037 and NIST SP 800-86 standards

TECHNICAL SKILLS

Cybersecurity: Network security architecture, vulnerability assessment, incident response, security policy development, VLAN segmentation, DMZ configuration

Security Tools: Security Onion, ELK Stack, Zed Attack Proxy (ZAP), Cisco Packet Tracer, Nessus (Basic)

Forensic Tools: Autopsy/Sleuth Kit, Volatility

Networking Technologies: Fortinet, TP-Link, Mikrotik, Cisco, VyOS, GNS3

Programming & Scripting: Python (Basic), C (Basic), MATLAB (Intermediate)

Operating Systems: Linux (Intermediate), Windows

Software & Tools: MS Office (Intermediate), Google Suite (Intermediate), Git

LANGUAGES

Spanish (Native) | **English** (Advanced)

PROFESSIONAL INTERESTS

Network Security: Passionate about designing robust network security architectures and implementing defense-in-depth strategies to protect organizational assets

Incident Detection & Response: Enthusiastic about developing and improving security monitoring capabilities and incident response procedures

LLMs at the Intersection of Cybersecurity: Interested in how large language models reshape offensive and defensive security landscapes, including the novel attack surfaces they introduce and their emerging role in autonomous network operations